Human-AI Collaboration in Programming: Analyzing the Acceptance of GitHub Copilot

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Abstract— Artificial intelligence tools like GitHub Copilot are increasingly integrated into software development, yet understanding the factors influencing their adoption remains underexplored. This study examines the determinants of software engineers' attitudes and intentions toward adopting GitHub Copilot from a human-centered perspective. By applying the Extended Technology Acceptance Model (TAM) and incorporating additional constructs, it identifies key drivers of adoption, including accuracy, intelligence, personal competency, job relevance, and social influence. A survey of 403 participants from Vietnam's programming community reveals that these factors significantly shape perceptions and behaviors, while perceived cost and risk exhibit minimal influence. The findings highlight the dual roles of perceived utility and social validation in fostering trust and acceptance of AI tools in programming. Accuracy and intelligence contribute to user confidence, while job relevance and social influence enhance perceived value. This research emphasizes the need for AI systems to prioritize contextual alignment with user needs, advancing both trust and efficiency in software development practices. The results have implications for designing AI tools like GitHub Copilot that integrate user-centered design principles to drive adoption and effectiveness. By addressing these factors, AI-powered programming tools can promote transparency, collaboration, and productivity, contributing to the broader discourse on human-AI collaboration in software engineering.

Keywords— Artificial intelligence , GitHub Copilot, Technology Acceptance Model (TAM), pair-programming, human-AI collaboration.

1. INTRODUCTION

A. Github Copilot

GitHub Copilot is an artificial intelligence (AI)-driven coding tool created to improve the coding process for developers by offering quicker and more effective code generation. It assists in suggesting code snippets as you type, whether it's completing a line or proposing an entire block of code [1]. GitHub Copilot can be accessed across multiple platforms. Developers can utilize it directly within your Integrated Development Environment (IDE) for seamless coding assistance. Additionally, GitHub Mobile provides a convenient chat interface for Copilot, allowing we to access its features on the go. For those who prefer working from the command line, GitHub CLI offers integration with Copilot, enabling coding support directly from the terminal. Moreover, subscribers to Copilot En-terprise can access its functionalities directly on GitHub.com, further enhancing their coding experience.

Many researchers have also published various papers related to Github Copilot since its release. In the year 2022, Madi conducted a study on assessing the readability and visual appeal of code generated by Github Copilot [2]. The research team led by Mastropaolo investigated the efficacy of Github Copilot in code generation. They utilized Github Copilot to generate 892 Java code snip-pets based on original Java documents and the research team's descriptions to evaluate their similarity [3]. So-bania evaluated the effectiveness of GitHub Copilot compared to genetic programming on standard synthesis benchmarks, demonstrating similar performance and highlighting the current limitations of genetic program-ming in practical software development [4]. Multiple research studies have been undertaken by author groups such as Imai [5] and Moradi [6] to investigate the productivity of pair programming involving GitHub Co-pilot and developers.

B. Technology acceptance

The original format of TAM was founded on two variables identified by Davis (1989) as the determining elements in the adoption of information technology: Perceived Ease of Use and Perceived Usefulness [7]. The degree to which a user believes that the technology will bring about ben-efits is referred to as the perceived usefulness of the technology. A user's perception of how easy and con-venient it is to utilize a particular piece of technology is referred to as the device's perceived ease of use. Based on the TAM, consumers are more inclined to adopt and utilize a technology if they perceive it to be useful for their work or meeting their needs, and if they find it easy to use. Besides the two main variables, researchers can add additional variables, depending on the field, tech-nology or application that the research topic is interested in. Due to its simplicity and clarity, TAM is widely applied in research across various fields, from information technology applications to e-commerce systems and mobile applications.

C. AI adoption

In early 2024, Dahri and his team conducted research to extend the technology acceptance model for ChatGPT, supported by AI, to aid in self regulated learning and metacognition in education [8]. In the education sector, the TAM is employed to assess the adoption of ChatGPT. This is done by verifying TAM through the widespread use of the learning mechanism [9]. Mohamed and his colleagues, have undertaken a study on leadership styles and the acceptability of AI in university libraries within the context of higher education [10], utilizing an extended TAM in their research. A study on the psychological factors and readiness of university lecturer to adopt AI based assessment systems has been introduced by Shahid and his colleagues [11].

In the programming domain, several research studies have also discussed the acceptance of AI as a tool to support programmers. The research conducted by Zelin Pan and colleagues investigated various aspects influencing the acceptance of AI as programming help tools among university students [12]. The study examines the utilization of ChatGPT as a collaborative programming tool in the realm of education, specifically in the context of programming instruction [13].

D. Research objectives

Although the TAM has been widely used to study technology adoption behaviors in several fields, its applicability in the specific context of software developers using GitHub Copilot has not been thoroughly investigated. The objective of this study is to incorporate more factors into the original TAM model. The supplementary extended variables were chosen from numerous prior investigations on the acceptability of new technologies. Subsequently, we propose a Github Copilot acceptance model. Using that model, we will analyze the characteristics that have a beneficial influence on the intents and actions of software engineers when they make the decision to utilize the Github Copilot tool.

Hypothetically, we hypothesize that extended variables will have a positive impact on perceived usefulness, perceived ease of use, as well as intention and behavior in deciding to use Github Copilot of software developers. In total, we have 23 hypotheses related to the original 4 variables of the TAM and 8 expanded variables. We will present these hypotheses in detail in the research methods section.

This study focuses on understanding the factors influencing software developer adoption of GitHub Copilot. We collected data for our research from individuals who are either currently studying information technology or working as professionals in the field of

software development. There are a total of 43 questions given to users through an online survey. These questions are related to 12 variables in our proposed model. Indeed, we have collected a total of 403 responses, which were provided by individuals representing various universities and programming communities in Vietnam.

2. Research Methodology

A. Theoretical framework

According to the original fundamental TAM model, which includes the variables Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Attitude Towards Use (ATU), and Behavioral Intention to Use (BI). We have included supplementary variables that have been cited in numerous prior research. Next, we provide the Github Copilot adoption model depicted in the graphic below.

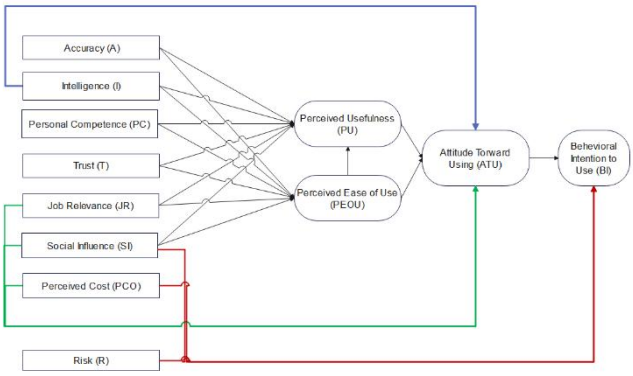

Fig. 1. Proposed Github Copilot acceptance model

The model also includes several supplementary variables that have been cited in prior research. These include Accuracy (A), Intelligence (I), Personal Competence (PC), Trust (T), Job Relevance (JR), Social Influence (SI), and Perceived Cost (PCO). These factors may influence the core TAM components of Perceived Usefulness, Perceived Ease of Use, Attitude Towards Use, and Behavioral Intention to Use the technology.

B. Hypothesis

TABLE I. LIST OF HYPOTHESIS

#	Hypothesis
H01a	The perceived accuracy of AI positively predicts the PU of Github Copilot.
H01b	The perceived accuracy of AI positively predicts the PEOU of Github Copilot.
H02a	The perceived intelligence of AI positively predicts the the PU of Github Copilot
H02b	The perceived intelligence of AI positively predicts the PEOU of Github Copilot
H02c	The perceived intelligence of AI positively predicts the attitude toward using Github Copilot
H03a	The user's personal competency positively predicts the PU of Github Copilot
H03b	The user's personal competency positively predicts the PEOU of Github Copilot
H04a	The level of trust that people have in AI positively predicts the PU of Github Copilot
H04b	The level of trust that people have in AI positively predicts the PEOU of Github Copilot
H05a	User's Job Relevance positively predicts the PU of Github Copilot
H05b	User's Job Relevance positively predicts the PEOU of Github Copilot
H05c	User's Job Relevance positively predicts the ATU Github Copilot
H06a	Social influence has a favorable correlation with the the PU of Github Copilot
H06b	Social influence has a favorable correlation with the PEOU of Github Copilot
H06c	Social influence has a favorable correlation with the ATU of Github Copilot
H06d	Social influence has a favorable correlation with the BI of Github Copilot
H07a	Perceived Cost positively predicts the attitude toward using Github Copilot
H07b	Perceived Cost positively predicts the behevioral intention to use Github Copilot
H08	Perceived Risk positively predicts the behevioral intention to use Github Copilot
H09	PU of Github Copilot positively predicts the ATU of Github Copilot
H10a	PEOU of Github Copilot positively predicts the PU of Github Copilot
H10b	PEOU of Github Copilot positively predicts the ATU of Github Copilot
H11	ATU of Github Copilot positively predicts the BI to use Github Copilot

C. Research instrument

The Likert scale is a measurement tool that presents a series of attitude-related statements. Respondents express their sentiments by selecting one of the replies, which range from strongly agree to strongly disagree (Batterton & Hale, 2017).

Based on previous research and the proposed model, we have developed a set of questions and utilized the Likert scale to create a questionnaire. This questionnaire will be used as a tool to support the research in this study.

TABLE II. LIST OF CONSTRUCT AND INDICATORS

Construct / indicators	Abbreviation
Accuracy [14]	**ACC**
I feel that GitHub Copilot accurately generates code snippets in accordance with the prompt's requirements.	ACC1
I feel GitHub Copilot automatically suggests accurate source code to complete the programming task.	ACC2
Intelligence [8], [15]	**INT**
Github Copilot demonstrates a high level of intelligence in assisting with my programming task.	INT1
Github Copilot's problem solving abilities impress me.	INT2
I perceive Github Copilot as highly intelligent in understanding and supporting my programming task.	INT3
Github Copilot's intelligence significantly contributes to its effectiveness in supporting my programming task.	INT4
Personal Competence [8]	**PC**
I believe I have the necessary skills to effectively use Github Copilot for programming.	PC1
I feel confident in my ability to navigate and use Github Copilot.	PC2
I'm proficient at tweaking Github Copilot to enhance my programming abilities.	PC3
I have the knowledge and abilities needed to make the most of Github Copilot for programming.	PC4
Trust [16]	**TRU**
I trust that Github Copilot can generate the code I need to complete the programming task	TRU1

I trust that my source code is protected when using Github Copilot	TRU2
I believe Github Copilot is a reliable tool to enhance my programming skills.	TRU3
Risk [16]	**RIS**
I'm concerned that using Github Copilot for programming will collect personal information from me	RIS1
I am concerned that using Github Copilot for programming will reveal confidential technological information about my source code and product	RIS2
I am concerned that Github Copilot use my source code for other purposes without my authorization	RIS3
Job Relevant [17]	**JR**
In my job, the usage of a AI tool as GitHub Copilot is important	JR1
In my job, the usage of a AI tools as Github Copilot is relevant	JR2
Social Influence [8]	**SI**
The opinions of my peers influenced my decision to use Github Copilot for programming.	SI1
I consider the recommendations of my boss/instructor/tutor when deciding to use Github Copilot.	SI2
My friends' perceptions of Github Copilot impact my acceptance and utilization.	SI3
I am influenced by the experiences and feedback of others in my programming community regarding Github Copilot.	SI4
Guidance from programming experts played an important role in my choice to use Github Copilot for programming.	SI5
Perceived Cost [18]	**PCO**
I feel that the price for Github Copilot is very inexpensive	PCO1
In my opinion, Github Copilot is expensive	PCO2
If I use Github Copilot long term, it will cost me a lot of money	PCO3
Perceived Usefulness [17]	**PU**
Using GitHub Copilot will increase my productivity in programming tasks	PU1
Using GitHub Copilot will enhance the quality of my work in programming tasks	PU2
Using GitHub Copilot will help reduce the time and effort required for programming tasks	PU3
Using GitHub Copilot to optimize the process of searching and implementing code solutions	PU4
Using GitHub Copilot to help me explore new programming techniques and methods	PU5
Perceived Ease of Use [19]	**PEOU**
Learning to use Github Copilot would be easy for me	PEOU1
It would be easy for me to become skillful at using Github Copilot	PEOU2
Interacting with Github Copilot would not require a lot of my mental effort	PEOU3
Attitude Toward Using [20], [8]	**ATU**
I have a positive attitude toward using Github Copilot for programming process.	ATU1
I think Using Github Coplilot is a smart way to get things done	ATU2
I hold a constructive attitude regarding Github Copilot impact on my programming skills	ATU3
I think that using Github Copilot is a good idea	ATU4
My attitude toward using Github Copilot is optimistic and encouraging.	ATU5
Behevioral Intention to Use [20], [8]	**BI**
I plan to use Github Copilot in the future in developing software applications	BI1
I predict that I will use Github Copilot in software development	BI2
I have strong intentions to continue using Github Copilot to learn programming	BI3
My intention is to regularly use Github Copilot as a valuable programming tool.	BI4

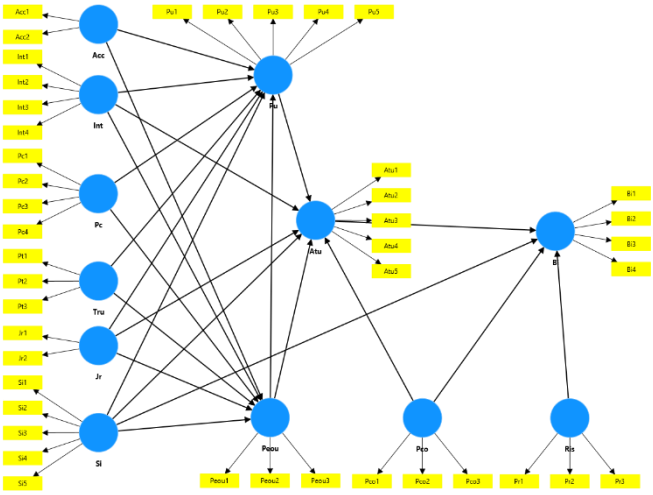

Fig. 2. Model Structure

D. Data collection

This study used a survey-based method to collect data for the purpose of testing the hypothesis. Participants were selected from three different sources in Vietnam. All of these participants were currently engaged in academic or professional activities in the field of software development, making them well suited for inclusion in our study. We obtained a total of 403 complete responses, all of which were valid and used for this study.

TABLE III. DEMOGRAPHICS OF PARTICIPANT

Items	Category	f	p (%)
Gender	Male	355	88.09
	Female	48	11.91
Age	Less than 18	10	2.48
	18 - 24	342	84.86

	25 - 34	41	10.17
	35 - 44	8	1.99
	45 and above	2	0.50
Programming Experience (Year)	One year or less	86	21.34
	1-2	342	84.86
	2-5	41	10.17
	More than 5	8	1.99
Level of interaction with AI tools	Do not use	9	2.23
	Occasionally use	150	37.22
	Regularly use	211	52.36
	Use specifically for daily tasks	33	8.19

f = Frequency; p (%) = Percentage of respondents

3. Data Analysis and Results

A. Model Measurement Analysis

We first used SmartPLS, a popular tool in data analysis, to calculate analytical values based on the PLS-SEM algorithm. After analyzing, we have the Outer Loading (LF) values of each indicator in Table IV, we have compiled a list of variables that are suitable for analysis. Variables are considered reliable when their Outer Loading values meet or exceed the threshold of 0.7, indicating a strong relationship with the corresponding latent variables. The PCO (Perceived Cost) variables are not considered reliable and suitable for further analysis, as their Outer Loading values are below the 0.7 cutoff (PCO1: 0.680, PCO2: 0.515, PCO3: 0.581).

TABLE IV. Results of Convergent Validity

Variable	CR	AVE	α
ACC	0.939	0.884	0.869
ATU	0.938	0.751	0.917
BI	0.906	0.706	0.861
INT	0.92	0.742	0.884
JR	0.909	0.833	0.799
PC	0.916	0.731	0.877
PCO	0.620	0.355	0.279
PEOU	0.896	0.742	0.826
PU	0.943	0.768	0.924
RIS	0.943	0.846	0.909
SI	0.903	0.652	0.868
TRU	0.911	0.773	0.854

α = Cronbach's alpha, AVE = Average variance extracted, CR = Composite reliability

Based on the reliability metrics of Cronbach's Alpha, Composite Reliability, and Average Variance Extracted, the variables ACC, ATU, BI, INT, JR, PC, PEOU, PU, RIS, SI, and TRU have been deemed acceptable. Only the PCO variable does not meet the necessary standards for reliability and validity, leading to its rejection. This comprehensive evaluation guarantees that the remaining variables are dependable and accurate for further analysis in comprehending the influence of GitHub Copilot on programming tasks.

Discriminant validity is shown when measures of different constructs that are not expected to be closely related are actually found to have low correlation with each other. The discriminant validity was assessed using HTMT (Heterotrait-Monotrait Ratio), which is considered to be more accurate than the Fornell-Larcker Criterion and Cross Loadings. The HTMT criteria indicate that if the HTMT value between two constructs is below 0.9, it suggests that discriminant validity has been adequately met. If the HTMT value is greater than 0.9, it suggests the presence of multicollinearity, which may impact the interpretation of the analysis results. Overall, the HTMT values for all constructs are below the critical threshold of 0.9, confirming that discriminant validity is adequately met. This validation supports the robustness of the measurement model and enhances the reliability of subsequent structural model analyses.

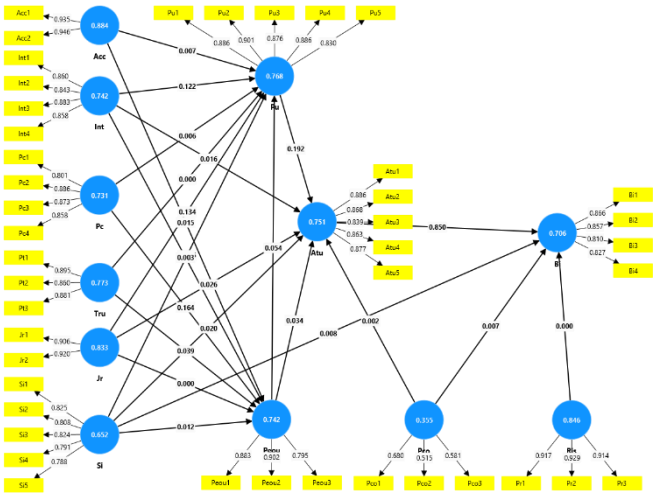

Fig. 3. Structural model with path coefficients

B. Hypothesis Testing

Based on the analyses in section 3.1, we decided to remove the variable PCO and the related hypotheses from this study. We then conducted Bootstrap calculations and used SmartPLS to test the remaining hypotheses.

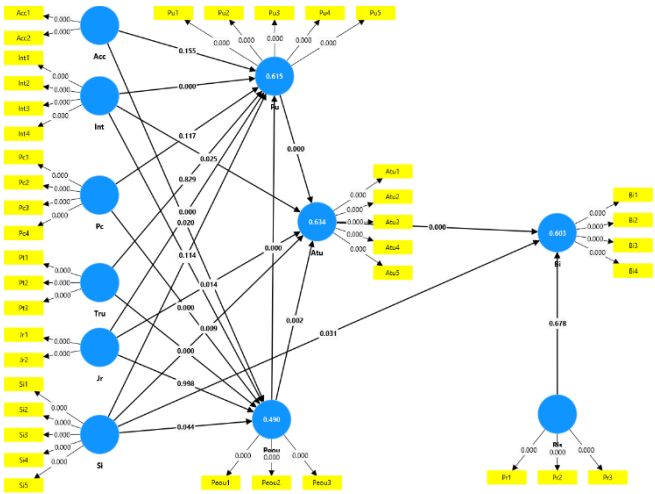

Fig. 4. Bootstrapping results showing p-values for hypothesis testing

TABLE V. Results of Testing Hypotheses

H	Path	t	p	I
H01a	ACC → PU	1.424	0.155	NS
H01b	ACC → PEOU	2.336	0.020	S
H02a	INT → PU	6.206	0.000	S
H02b	INT → PEOU	1.118	0.264	NS
H02c	INT → ATU	2.239	0.025	S
H03a	PC → PU	1.568	0.117	NS
H03b	PC → PEOU	6.67	0.000	S
H04a	TRU → PU	0.216	0.829	NS

H04b	TRU → PEOU	3.562	0.000	S
H05a	JR → PU	7.255	0.000	S
H05b	JR → PEOU	0.003	0.998	NS
H05c	JR → ATU	2.46	0.014	S
H06a	SI → PU	1.582	0.114	NS
H06b	SI → PEOU	2.014	0.044	S
H06c	SI → ATU	2.614	0.009	S
H06d	SI → BI	2.154	0.031	S
H08	RIS → BI	0.415	0.678	NS
H09	PU → ATU	7.362	0.000	S
H10a	PEOU → PU	3.95	0.000	S
H10b	PEOU → ATU	3.089	0.002	S
H11	ATU → BI	19.067	0.000	S

H (Hypothesis), t (t-value), p (p-value), I (Influence), S (Supported), NS(Not Supported)

The key findings from the analysis are: H01b, H02a, H02c, H03b, H04b, H05a, H05c, H06b, H06c, H06d, H09, H10a, and H10b were supported, indicating significant positive relationships. However, H01a, H02b, H03a, H04a, H05b, H06a, and H08 were not supported, showing insignificant relationships.

Perceptions of GitHub Copilot's accuracy positively impacted its perceived ease of use, but not its perceived usefulness. Perceived intelligence positively predicted usefulness and attitude, but not ease of use. Personal competence and trust in AI had positive effects on ease of use, but not usefulness. Job relevance strongly predicted usefulness and attitude. Social influence had favorable correlations with ease of use, attitude, and behavioral intention, but not usefulness. Perceived risk did not significantly impact behavioral intention. Consistent with TAM, perceived usefulness and ease of use were positive predictors of attitude, which then positively predicted behavioral intention to use GitHub Copilot.

C. Confirmation of research model

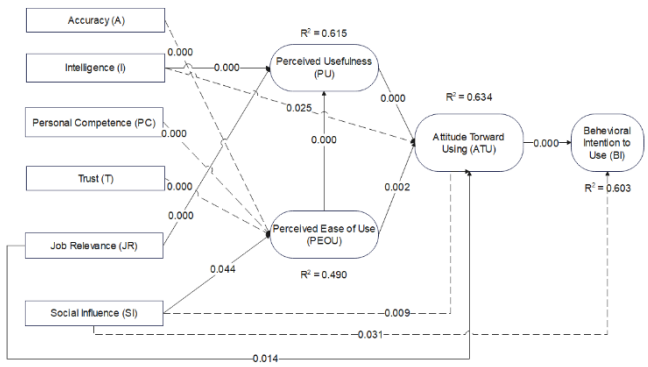

Fig. 5. Final structural model with significant relationships only, R^2 = coefficient of determination

TABLE VI. MODEL FITNESS SCORE -R2 OF THE ENDOGENOUS LATENT VARIABLES

Variables	R-square	R-square adjusted
ATU	0.634	0.630
BI	0.603	0.601
PEOU	0.490	0.484
PU	0.615	0.610

The model fitness score in Table VI, represented by the R-square (R^2) values of the endogenous latent variables, indicates the proportion of variance in these variables explained by the model. For the variable Attitude Toward Using (ATU), the R^2 is 0.634, meaning that 63.4% of the variance in ATU is explained by the model, with an adjusted R^2 of 0.630. Behavioral Intention (BI) has an R^2 of 0.603 and an adjusted R^2 of 0.601, indicating that 60.1% of its variance is accounted for by the model. Perceived Ease of Use (PEOU) shows an R^2 of 0.490 and an adjusted R^2 of 0.484, suggesting that 49% of the variance in PEOU is explained. Lastly, Perceived Usefulness (PU) has an R^2 of 0.615 and an adjusted R^2 of 0.610, meaning that 61.5% of the variance in PU is explained by the model. The R^2 values indicate that ATU, BI, and PU have a high level of explanatory power, whereas PEOU has a moderate level of explanatory power.

4. DISCUSSION

With data from 403 surveys, we conducted the analysis steps and got the following results: For Github Copilot's Perceived Usefulness (PU), the extended variables added to the TAM model that have a positive impact on it are the tool's intelligence level and the user's work relevance. This shows that depending on the user's work purpose, as well as the perception of the intelligence of this AI tool will affect the user in evaluating whether Github Copilot is useful or not. The value of Github Copilot's PEOU is significantly influenced by the accuracy of AI, the personal abilities and beliefs of the users, and the societal effect. Users are concerned about the accuracy of the findings provided by Github Copilot during programming. Furthermore, individuals possess distinct personal aptitude and societal impact that can influence their perceptions regarding the ease or complexity of utilizing Github Copilot Besides Github Copilot's Perceived Usefulness (PU) and Perceived Ease of Use (PEOU), the additional factors that influence the attitude towards utilizing Github Copilot are AI intelligence, user work relevancy, and social influence. The intention to utilize Github Copilot is positively influenced by the Attitude Towards utilize (ATU), and the variable of expansion represents the influence of society. The perceived risk of using GitHub Copilot does not appear to significantly affect attitudes and behavioral intentions towards it, in relation to the extended variable. Although these characteristics may still hold some importance for specific users, they do not play the primary role in deciding the adoption of Github Copilot. In addition, perceived cost was removed from the original proposed model because survey data showed that it was not suitable for the extended TAM model.

5. CONCLUSION

We have completed the analysis steps in this project. The goal of this study is to analyze programming practitioners' attitudes and intentions toward using Github Copilot. We expanded the original TAM model and added variables about AI accuracy, intelligence; personal capacity, trust, and suitability for the user's job; social influence and perceived costs were also added to the model for testing.

In our investigation, the individuals who participated in the survey were programming practitioners from Vietnam, including both students studying information technology and professionals working in software development. This could potentially affect the comprehensiveness of the research. Each nation will possess distinct cultural attributes, varying degrees of affluence, diverse economic magnitudes, and distinctive technological frameworks. Therefore, it is plausible that the findings of this study may not be relevant to programming practitioners in different geographical areas, where other factors might have a greater impact on the adoption of Github Copilot.

Although we proposed to add 8 expanded variables to the TAM model, 6 of them actually have an impact and are retained in the model after analysis. The number of these extension variables is still limited, there may be some other factors that have an impact on the adoption of Github Copilot by programming practitioners.

Future research directions include: Increase the number of variables that extend to the Github Copilot adoption model, which can include factors such as the software development environment, project complexity, and level of interaction between developers. Collect surveys from many countries to get a more comprehensive and diverse view of Github Copilot adoption and usage. This will help the research better understand how cultural and work environment differences influence the use of this tool. Conduct similar research for Github Copilot adoption compared with other

AI-powered programming tools, these tools could be ChatGPT, TabNine, Gemini, and other programming-enabled AI tools. Compare the adoption and effectiveness of these tools to determine the critical factors driving their success. These research directions will help expand understanding and improve the use of AI tools to support programming, thereby improving efficiency and quality in software development.

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